

Q1 - 24 June - Shift 1

If the sum of the squares of the reciprocals of the roots α and β of the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6(\alpha^3 + \beta^3)^2$ is equal to :

- (A) 18 (B) 24
(C) 36 (D) 96

Space for your notes:

Q2 - 24 June - Shift 2

The sum of all the real roots of the equation $(e^{2x} - 4)(6e^{2x} - 5e^x + 1) = 0$ is

- (A) $\log_e 3$ (B) $-\log_e 3$
(C) $\log_e 6$ (D) $-\log_e 6$

Space for your notes:

Q3 - 25 June - Shift 1

For a natural number n , let $a_n = 19^n - 12^n$. Then,

the value of $\frac{31\alpha_9 - \alpha_{10}}{57\alpha_8}$ is

Space for your notes:

Q4 - 25 June - Shift 2

Let $a, b \in \mathbb{R}$ be such that the equation $ax^2 - 2bx + 15 = 0$ has a repeated root α . If α and β are the roots of the equation $x^2 - 2bx + 21 = 0$, then $\alpha^2 + \beta^2$ is equal to:

- (A) 37 (B) 58
(C) 68 (D) 92

Space for your notes:

Q5 - 26 June - Shift 1

The sum of the cubes of all the roots of the equation $x^4 - 3x^3 - 2x^2 + 3x + 1 = 10$ is _____.

Space for your notes:

Q6 - 27 June - Shift 1

If the sum of all the roots of the equation $e^{2x} - 11e^x - 45e^{-x} + \frac{81}{2} = 0$ is $\log_e P$, then p is equal to _____.

Space for your notes:

Q7 - 27 June - Shift 2

Let α, β be the roots of the equation $x^2 - 4\lambda x + 5 = 0$ and α, γ be the roots of the equation $x^2 - (3\sqrt{2} + 2\sqrt{3})x + 7 + 3\lambda\sqrt{3} = 0$.

Space for your notes:

If $\beta + \gamma = 3\sqrt{2}$, then $(\alpha + 2\beta + \gamma)^2$ is equal to :

Q8 - 28 June - Shift 1

The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^x + 1 = 0$ is _____.

Space for your notes:

Q9 - 28 June - Shift 2

Questions

MathonGo

Let $f(x)$ be a quadratic polynomial such that $f(-2) + f(3) = 0$. If one of the roots of $f(x) = 0$ is -1 , then the sum of the roots of $f(x) = 0$ is equal to :

Space for your notes:

(A) $\frac{11}{3}$

(B) $\frac{7}{3}$

(C) $\frac{13}{3}$

(D) $\frac{14}{3}$

Q10 - 29 June - Shift 2

Let α be a root of the equation $1 + x^2 + x^4 = 0$.

Space for your notes:

Then the value of $\alpha^{1011} + \alpha^{2022} - \alpha^{3033}$ is equal to:

(A) 1

(B) α

(C) $1 + \alpha$

(D) $1 + 2\alpha$

#MathBoleTohMathonGo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

Answer Key

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

Q1 (B)

Q2 (B)

Q3 (4)

Q4 (B)

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

Q5 (36)

Q6 (45)

Q7 (98)

Q8 (2)

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

Q9 (A)

Q10 (A)

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

#MathBoleTohMathonGo

Q1 (B)

Here α, β roots of equation $3x^2 + \lambda x - 1 = 0$

$$\alpha + \beta = \frac{-\lambda}{3}, \alpha\beta = \frac{-1}{3}$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha^2\beta^2} = 15$$

$$\lambda^2 = 9$$

$$\text{Now } 6(\alpha^3 + \beta^3)^2 = 6\left((\alpha + \beta)\left((\alpha + \beta)^2 - 3\alpha\beta\right)\right)^2$$

$$= 6\left(\frac{\lambda^2}{9}\right)\left\{\frac{\lambda^2}{9} + 1\right\}^2 = 24$$

Q2 (B)

$$(e^{2x} - 4)(6e^{2x} - 3e^x - 2e^x + 1) = 0$$

$$(e^{2x} - 4)(3e^x - 1)(2e^x - 1) = 0$$

$$e^{2x} = 4 \text{ or } e^x = \frac{1}{3} \text{ or } e^x = \frac{1}{2}$$

$$\Rightarrow \text{sum of real roots} = \frac{1}{2} \ln 4 + \ln \frac{1}{3} + \ln \frac{1}{2}$$

$$= -\ln 3$$

Q3 (4)

$$a_n = 19^n - 12^n$$

$$\frac{31\alpha_9 - \alpha_{10}}{57\alpha_8} = \frac{31(19^9 - 12^9) - (19^{10} - 12^{10})}{57\alpha_8}$$

$$= \frac{19^9(31 - 19) - 12^9(31 - 12)}{57\alpha_8}$$

$$= \frac{19^9 \cdot 12 - 12^{10} \cdot 19}{57\alpha_8}$$

$$= \frac{12 \cdot 19(19^8 - 12^8)}{57\alpha_8} = 4$$

Q4 (B)

$$ax^2 - 2bx + 15 = 0$$

$$2\alpha = \frac{2b}{a}, \alpha^2 = \frac{15}{a}$$

$$\frac{\alpha}{2} = \frac{15}{2b}$$

$$\alpha = \frac{15}{b}$$

$$x^2 - 2bx + 21 = 0$$

$$\left(\frac{15}{b}\right)^2 - 2b\left(\frac{15}{b}\right) + 21 = 0$$

$$b^2 = 25$$

$$\alpha + \beta = 2b, \alpha\beta = 21$$

$$\alpha^2 + \beta^2 = 4b^2 - 42$$

$$= 58$$

Q5 (36)

Hints and Solutions

MathonGo

$$x^4 - 3x^3 - 2x^2 + 3x + 1 = 10$$

$x = 0$ is not the root of this equation so divide it by x^2

$$x^2 - 3x - 2 + \frac{3}{x} + \frac{1}{x^2} = 0$$

$$x^2 + \frac{1}{x^2} - 2 + 2 - 3\left(x - \frac{1}{x}\right) - 2 = 0$$

$$\left(x - \frac{1}{x}\right)^2 - 3\left(x - \frac{1}{x}\right) = 0$$

$$x - \frac{1}{x} = 0,$$

$$x - \frac{1}{x} = 3$$

$$x^2 - 1 = 0$$

$$x^2 - 3x - 1 = 0$$

$$x = \pm 1$$

$$\gamma + \delta = 3$$

$$\alpha = 1, \beta = -1$$

$$\gamma\delta = -1$$

$$\alpha^3 + \beta^3 + \gamma^3 + \delta^3$$

$$1 - 1 + (\gamma + \delta)((\gamma + \delta)^2 - 3\gamma\delta)$$

$$0 + 3(9 - 3(-1))$$

$$+ 3(12) = 36$$

Q6 (45)

#MathBoleTohMathonGo

To practice more chapter-wise JEE Main PYQs, [click here to download the MARKS app](#) from Playstore

$$e^{2x} - 11e^x - 45e^{-x} + \frac{81}{2} = 0]$$

$$(e^x)^3 - 11(e^x)^2 - 45 + \frac{81e^x}{2} = 0.]$$

$$e^x = t$$

$$2t^3 - 22t^2 + 81t - 90 = 0$$

$$t_1 t_2 t_3 = 45$$

$$e^{x_1} \cdot e^{x_2} \cdot e^{x_3} = 45$$

$$e^{x_1 + x_2 + x_3} = 45$$

$$\log_e e^{x_1 + x_2 + x_3} = \log_e 45$$

$$x_1 + x_2 + x_3 = \log_e 45$$

$$\log_e P = \log_e 45$$

$$P = 45$$

Q7 (98)

$$x^2 - 4\lambda x + 5 = 0 \left\{ \begin{array}{l} \alpha \\ \beta \end{array} \right.$$

$$x^2 - (3\sqrt{2} + 2\sqrt{3})x + (7 + 3\lambda\sqrt{3}) = 0 \left\{ \begin{array}{l} \alpha \\ \gamma \end{array} \right.$$

$$\alpha + \beta = 4\lambda$$

$$\alpha + \gamma = 3\sqrt{2} + 2\sqrt{3}$$

$$\beta + \lambda = 3\sqrt{2}$$

$$\alpha\gamma = 7 + 3\lambda\sqrt{3}$$

$$\therefore \alpha = 2\lambda + \sqrt{3}$$

$$\alpha\beta = 5$$

$$\beta = 2\lambda - \sqrt{3}$$

$$4\lambda^2 = 8 \Rightarrow \lambda = \sqrt{2}$$

$$\therefore (\alpha + 2\beta + \lambda)^2 = (4\alpha + 3\sqrt{2})^2 = (7\sqrt{2})^2 = 98$$

Q8 (2)

$$e^{4x} + 4e^{3x} - 58e^{2x} + 4e^x + 1 = 0$$

$$\text{Let } f(x) = e^{2x} \left(e^{2x} + \frac{1}{e^{2x}} + 4 \left(e^x + \frac{1}{e^x} \right) - 58 \right)$$

$$e^x + \frac{1}{e^x}$$

$$\text{Let } h(t) = t^2 + 4t - 58 = 0$$

$$t = \frac{-4 \pm \sqrt{16 + 4 \cdot 58}}{2}$$

$$\frac{-4 \pm 2\sqrt{62}}{2}$$

$$t_1 = -2 + 2\sqrt{62}$$

$$t_2 = -2 - 2\sqrt{62} \text{ (not possible)}$$

$$t \geq 2$$

$$e^x + \frac{1}{e^x} = -2 + 2\sqrt{62}$$

$$e^{2x} - (-2 + 2\sqrt{62})e^x + 1 = 0$$

$$(-2 + 2\sqrt{62}) - 4$$

$$4 + 4 \cdot 62 - 8\sqrt{62} - 4$$

$$248 - 8\sqrt{62} > 0$$

$$\frac{-b}{2a} > 0$$

both roots are positive

2 real roots

Q9 (A)

$$f(-2) + f(3) = 0$$

$$f(x) = (x + 1)(ax + b)$$

$$f(-2) + f(3) = -1(-2a + b) + 4(3a + b) = 0$$

$$2a - b + 12a + 4b = 0$$

$$14a + 3b = 0$$

$$\frac{-b}{a} = \frac{14}{3}$$

$$\text{Sum of roots} = \left(-1 + \frac{-b}{a}\right) = -1 + \frac{14}{3} = \frac{11}{3}$$

Q10 (A)

$$x^4 + x^2 + 1 = 0$$

$$\Rightarrow (x^2 + x + 1)(x^2 - x + 1) = 0$$

$$\Rightarrow x = \pm \omega, \pm \omega^2 \text{ where } \omega = 1^{1/3} \text{ and imaginary.}$$

$$\text{So } \alpha^{1011} + \alpha^{2022} - \alpha^{3033} = 1 + 1 - 1 = 1$$